

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 - Low (Pseudo-code Writing)

Course Code:	CSA0501	Course Title:	Programming in Java
CO Assessed:	CO1 - Precision (BL3,BL4)	Assessment:	Assessment Tool 3 - Pseudo-code Writing
Weightage:		Total Marks:	100
CO1 Statement:	Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)		
Student Name:	S. Aaradhana	Reg. No.:	1921260021
Section Batch:	/ SLOT B	Date:	27.07.26

Code of Conduct

I S. Aaradhana (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- This test contains 10 questions, each carrying 10 marks. Total: 100 Marks.
- No negative marking. Un answered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

20

25

16

10

20

91
100

88

CO-1 ASSESMENT TOOL - 3

NAME: S. Ananthana
RG. NO.: 1921260021
SUBJECT: Java
SUB CODE: CSA0901

PSEUDO-CODE WRITING

1. Student Management System

Write pseudo code to design a student class with attributes (Name, RollNo, Marks) and method to:

- Accept student details
- Calculate average marks
- Display result (Pass/Fail)

Apply encapsulation and data hiding by restricting direct access to attributes.

OOP Concepts: Encapsulation | Data Hiding | Classes and Object.

START

CLASS Student

PRIVATE name, rollno, marks1, marks2, marks3

METHOD acceptDetails()

INPUT name, rollNo

INPUT marks1, marks2, marks3

END METHOD

METHOD Calculate displayResult()

average = calculateAverage()

IF average $\geq$ 50 THEN

PRINT "Pass"

ELSE

PRINT "Fail"

END IF

END METHOD

END CLASS

CREATE Student s

CALL s.acceptDetails()

CALL s.displayResult()

2. Library Book Management

Design a book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:

- Issue a book (change availability status)
- Return a book
- Display book details using ^{getter} methods.

Demonstrate object creation for at least 3 books and simulate issue/return operations.

OOP Concepts: Encapsulation | Getters and setters | objects.

START

CLASS BOOK

PRIVATE title, author, ISBN, isAvailable

METHOD setDetails (t, a, i)

title = t

author = a

ISBN = i

isAvailable = TRUE

END METHOD

METHOD issueBook()

IF isAvailable = TRUE THEN

isAvailable = FALSE

PRINT "Book Issued"

ELSE

PRINT "Not available"

END IF

END METHOD

METHOD returnBook()

isAvailable = TRUE

PRINT "Book Returned"

END METHOD

METHOD display()

PRINT title, author, ISBN, isAvailable

END METHOD

END CLASS

CREATE BOOK b1, b2, b3

SET details for all books

CALL b1.issueBook()

CALL b1.returnBook()

CALL b1.display()

CALL b2.display()

CALL b3.display()

STOP

3. Employee Salary Calculation

Write pseudocode for an Employee class with encapsulation attributes (empld, name, basicPay, designation). Implement to

- calculate gross salary (basic + HRA + DA)
- calculate net salary after deduction (PF, tax)
- display pay slip.

Apply data hiding so salary components are not directly accessible OOP concepts: Encapsulation | Data Hiding | Methods.

START

CLASS Employee

PRIVATE empld, name, basicPay, designation

METHOD accept()

INPUT empld, name, designation, basic pay

END METHOD

METHOD calculateGross()

HRA = basicPay * 0.20

DA = basicPay * 0.10

gross = basicPay + HRA + DA

RETURN gross

END METHOD

METHOD calculateNet()

gross = calculateGross()

PF = basicPay * 0.12

tax = gross * 0.05

net = gross - PF - tax

RETURN net

END METHOD

METHOD displayPaySlip()

PRINT empld, name

PRINT "Gross Salary =", calculateGross()

PRINT "Net Salary =", calculateNet()

END METHOD

END CLASS

CREATE Employee e

CALL e.accept()

CALL e.displayPaySlip()

STOP.